Homework 1(c) - Data Visualization (Quantitative Data)

1 Histogram

The following data represents the age distribution of 50 students:

- 15-20 years: 12 students
- 21-25 years: 18 students
- 26-30 years: 10 students
- 31-35 years: 6 students
- 36-40 years: 4 students

1.1 Python Code for Histogram

```
import matplotlib.pyplot as plt

age_groups = ['15-20', '21-25', '26-30', '31-35', '36-40'] students = [12, 18, 10, 6, 4]

plt.bar(age_groups, students, color='blue', edgecolor='black') plt.xlabel('Age Groups') plt.
```

2 Frequency Polygon

The following data represents hours studied by students:

Hours Studied	Frequency
0-5	8
5-10	12
10-15	15
15-20	10
20-25	5

2.1 Python Code for Frequency Polygon

```
import matplotlib.pyplot as plt

classes = [(0, 5), (5, 10), (10, 15), (15, 20), (20, 25)] frequencies = [8, 12, 15, 10, 5] m

plt.plot(midpoints, frequencies, marker='o', linestyle='-', color='b', label='Frequency Poly
```

3 Frequency Curve

3.1 Python Code for Frequency Curve

```
import numpy as np import matplotlib.pyplot as plt from scipy.stats import norm

data = [0, 5, 10, 15, 20] frequencies = [8, 12, 15, 10, 5] mu, std = norm.fit(data)

x = np.linspace(min(data), max(data), 100) p = norm.pdf(x, mu, std) plt.plot(x, p, 'k', line
```

4 Ogive

The cumulative frequency distribution for hours studied is:

Hours Studied	Cumulative Frequency
0-5	8
5-10	20
10-15	35
15-20	45
20-25	50

4.1 Python Code for Ogive

```
import matplotlib.pyplot as plt import numpy as np

classes = [(0, 5), (5, 10), (10, 15), (15, 20), (20, 25)] frequencies = [8, 12, 15, 10, 5] c

plt.plot(upper_boundaries, cumulative_frequencies, marker='o', linestyle='-', color='r', lab
```